

# Fernando Hernandez Cantu

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[fhc-portfolio.vercel.app](https://fhc-portfolio.vercel.app)

## EDUCATION

Monterrey Institute of Technology and Higher Education (ITESM)

Aug 2024 – Jun 2028

Bachelor of Computer Engineering (GPA: 95/100)

Monterrey, N.L.

## EXPERIENCE

### Epicor

Dec 2025 – Jun 2026

Software Engineering Intern

Monterrey, N.L.

- Contributed to **Prism**, Epicor's new multi-agent AI assistant for the **Eclipse ERP**, presented as a beta at the company's international **Insights** conference.
- Built **Python** sub-agents for inventory queries, business summaries, and **Solar** ERP UI actions, using **OpenAI** calls.
- Created **C#/.NET** endpoints exposing ERP data and business logic as agent tools.
- Integrated the **Angular** chatbot UI into the **Java**-based **Solar** ERP application.

### Distritotec

Aug 2025 – Feb 2026

Software Engineering Intern

Monterrey, N.L.

- Engineered an end-to-end pipeline tracking 500+ people/day for crowd estimation and person registration at ITESM's Central Park, using **YOLO** and **ByteTrack** for multi-object tracking.
- Trained a **PyTorch** classification model using **ResNet18** and **Attention Pooling** to extract temporal embeddings for child vs. adult inference via an **MLP**, achieving 90% accuracy.
- Built scalable backend services using **PostgreSQL**, **PostGIS**, and **Docker** within a microservices architecture deployed on **Azure** cloud infrastructure.

## PROJECTS

RoboCup @Home | Vision & Manipulation Lead, RoBorregos

May 2025 – Present

- 9th Place** at **RoboCup 2026 - Incheon, Korea** and **1st Place** at the **Mexican Robotics Tournament 2026**.
- Engineered an **ML pipeline (FastAPI, Docker)** automating dataset labeling with **Grounding DINO + SAM**, synthetic dataset generation, and multi-GPU **YOLO** training from a web UI.
- Designed a computer vision architecture (**Docker, gRPC, ROS2**) integrating face recognition, pose estimation, object detection, and **Moondream** modules for real-time robot tasks.
- Developed an adaptive grasp-pose estimator from **ZED2** point clouds for a **MoveIt/ROS2** pipeline.

RoboWeb | Full-Stack Developer for RoBorregos Web Projects

May 2025 – Present

- Candidates Dashboard:** Created a **T3 Stack** platform to manage the team's annual recruiting competition for 100+ participants and staff, featuring a live scoreboard, automated round scheduling, and a judging system.
- Dashboard-Tec:** Developed a **Next.js/tRPC/PostgreSQL** community platform for ITESM events, building the event calendar and admin panels.
- SWE Training:** Launched a platform to manage LeetCode resources and leaderboards with **T3 Stack**, supporting 35+ students in solving 650+ technical problems.

Smart Parking IoT | Software Developer for the ITESM "Expo Ingenierías"

Sep 2025 – Dec 2025

- Implemented a smart-parking system with automatic **license plate recognition** in **Python/FastAPI**.
- Integrated **ESP32/ESP32-CAM** sensors with a **Node.js** server over **MQTT**, persisting data to **PostgreSQL** via **Prisma**.
- Built a real-time dashboard to monitor parking availability and live sensor data.

RoboCup Rescue Maze Jr. | Team Captain

Nov 2024 – May 2025

- Achieved **1st place** at Regionals and **2nd place** at the **Mexican Robotics Tournament**.
- Developed a **TensorFlow** vision pipeline for victim detection on a **Jetson Nano-ESP32** communication system.
- Architected an autonomous navigation system in **C++** using **PID** control and **DFS/Dijkstra** path planning.

## TECHNICAL SKILLS

**Languages:** C++, Python (3y) · TypeScript/JavaScript, SQL (2y) · Java, C# (1y)

**Frameworks:** React, Next.js, NestJS, Node.js, NextAuth.js, .NET, gRPC, Tailwind, Prisma, tRPC, FastAPI, TensorFlow, PyTorch, ROS2

**Technologies:** Git, Linux, Docker, Azure, PostgreSQL, MySQL, OpenCV, YOLO, Gazebo, Arduino, SolidWorks

## AWARDS & LANGUAGES

11th gen. "Líderes del Mañana" ITESM - Full Scholarship · Bronze Medal National Math Olympiad ONMB (2024) · 2nd Place National Academic Festival DGTI (2023) · Bronze Medal Math Olympiad ONMAPS (2021)

**Languages:** Spanish (Native), English (C1)